

A Leakage-Aware and Auditable Framework Prioritizes Class I HDAC Inhibitors for Pan-Cancer Transcriptomic Reversal

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Abstract

Background: Transcriptomic reversal can prioritize compounds whose perturbational expression profiles oppose disease-associated programs, but reliability depends on molecular-identity control, leakage-aware evaluation, chemical-space assessment, and a clear distinction between predicted and measured signatures. We evaluated a leakage-aware and auditable framework for pan-cancer transcriptomic-reversal analysis and candidate prioritization, with molecular-representation comparison treated as one component of the evidence audit.

Methods: A dual-stream atom-token/fingerprint model and a strong conventional fingerprint multilayer perceptron (MLP) comparator were trained on 55,695 quality-controlled LINCS L1000 Level 5 signatures. Performance was evaluated using drug-cell pair, leave-drug-out, leave-cell-line-out, scaffold, and corrected annotation-defined HDAC holdouts with repeated random seeds. Both chemical-structure-only models screened 28,477 compounds against disease signatures from 22 TCGA cancer types. Candidate stability was primary across 48 split-model-seed-metric configurations; a legacy metric expanded this to 72 configurations only as sensitivity analysis. Predicted reversal was compared with measured LINCS profiles using official perturbagen, dose, time, cell-line, and quality annotations and crossed candidate-cancer resampling. Candidate identity, formal HDAC enrichment, structural-neighbor exposure, reversal-associated networks, DepMap dependencies, crystallographic redocking, a zinc-chelation decoy, and receptor sensitivity were audited after candidate tiers were frozen.

Results: The dual-stream architecture provided no measurable performance gain over the strong conventional fingerprint MLP comparator. The fingerprint MLP was slightly better on average in the pair, leave-drug, leave-cell-line, and scaffold settings, while performance was comparable in the corrected annotation-defined HDAC holdout (1,856 profiles from 30 unseen structures): mean Pearson correlations were 0.378 for the dual-stream model and 0.379 for the fingerprint MLP, and mean Spearman correlations were 0.345 and 0.344, respectively. Strict candidate-level leave-drug evaluation was available for Mocetinostat and PCI-24781 and did not favor the dual-stream model. Signed wTRS enriched the explicitly annotated class I HDAC subset at the fixed revision cutoffs of the top 0.5%, 1%, 5%, and 10% of the library (fold enrichments 30.6, 19.2, 8.46, and 4.62; all FDR $< 10^{-4}$), whereas the co-primary Spearman reversal metric did not. Because signed wTRS is sensitive to perturbational amplitude whereas Spearman correlation is scale invariant, the enrichment is restricted to signed wTRS and may partly reflect response magnitude. Across eight compounds with official high-quality measured LINCS profiles and 22 cancer signatures, predicted and measured reversal were concordant for both models

(Spearman 0.640–0.830; crossed-bootstrap lower 95% limits 0.297–0.653, depending on model and metric). Mocetinostat was retained as the core candidate, NCH-51 as secondary, and TC-H-106 as exploratory. RG2833 lacked a measured LINCS signature, whereas Tianeptinaline/BG-1010 had an identity conflict and was excluded from primary inference. DepMap supported HDAC3, rather than HDAC1, as the dominant pan-cancer class I HDAC dependency. Zinc-aware redocking recovered the crystallographic Vorinostat pose, but a designed decoy showed that favorable docking scores alone did not establish zinc-chelating geometry.

Conclusions: Rigorous split design, leakage and structural-proximity auditing, and calibrated evidence layers support hypothesis-generating candidate prioritization. The present benchmarks do not justify the additional complexity and computational cost of the dual-stream representation. The framework separates prediction generalization, predicted–measured transcriptomic concordance, biological context, and structural sensitivity without implying metric-independent class enrichment, direct target engagement, or therapeutic efficacy.

Keywords: transcriptomic reversal; drug repurposing; LINCS L1000; TCGA; histone deacetylase; leakage-aware evaluation; auditability; applicability assessment.

1 Background

Large-scale cancer profiling has documented substantial genomic, transcriptomic, and epigenetic heterogeneity within and across malignancies [1–3]. Although these resources have enabled the discovery of lineage-specific drivers and shared molecular vulnerabilities, translating high-dimensional tumor profiles into experimentally tractable therapeutic hypotheses remains difficult. Drug repurposing offers a pragmatic route for hypothesis generation because compounds with existing chemical or pharmacological information can be considered in new disease contexts more rapidly than entirely new chemical entities [4, 5].

Transcriptomic reversal is one computational principle for such prioritization. A perturbation is ranked more highly when its expression signature is directionally opposite to a disease-associated signature [6]. The LINCS L1000 resource provides large-scale measured perturbational profiles across compounds, cellular backgrounds, doses, and times [8], while oncology screens illustrate the complementary value of direct viability measurements [7]. These endpoints are not interchangeable. Reversal can identify a compound-associated expression pattern worth testing, but it does not

demonstrate binding, growth inhibition, a therapeutic window, or clinical benefit.

An important modeling question is how small molecules should be represented. Molecular graphs, graph-derived atom descriptors, fingerprints, and learned sequence representations encode different aspects of structure [9–14]. However, systematic benchmarking has found limited gains from representation-learning models across many molecular-property datasets and has emphasized the influence of dataset quality, split design, chemical-space coverage, and evaluation choices [37]. The submitted model combined an atom-token Transformer stream with a Morgan-fingerprint stream. Its atom stream uses padded element-identity tokens and does not perform bond-conditioned message passing; we therefore describe it precisely as an atom-token encoder rather than as a bond-aware graph neural network.

Related frameworks have different objectives and input assumptions. DeepPurpose provides reusable encoders for drug–target interaction prediction [21]; AttentiveFP uses graph attention for molecular property learning [22]; DeepCE predicts chemical-perturbation expression profiles using drug and cellular context [23]; and ChemCPA models compositional perturbation responses [24]. The present work does not claim a new neural primitive or report proxy numbers as if these published systems had been reproduced under their native protocols. The representation comparison is an informative component rather than the principal methodological advance. The primary contribution is a leakage-aware and auditable framework that connects pan-cancer reversal to entity-aware splits, chemical-space and structural-proximity assessment, official-condition measured profiles, frozen candidate tiers, and explicit evidence labels. The fingerprint-only MLP is treated as a strong conventional comparator trained and evaluated with the identical target matrix and split membership.

Class I histone deacetylase (HDAC) inhibition is already well established in oncology research [34–36]. Consequently, recurrent HDAC enrichment is not presented as discovery of an unknown drug class. The revised question is whether a leakage-aware reversal pipeline can prioritize specific HDAC-associated hypotheses, distinguish measured from prediction-only support, and identify where its conclusions remain sensitive to the reversal metric, structural neighbors, or analytical choices. Candidate-specific central nervous system exposure may motivate later experiments [30, 31], but brain penetrance is not inferred from the present data and is not used as the primary ranking criterion.

The framework addressed three questions. First, does the added dual-stream representation complexity provide a measurable advantage over the strong conventional fingerprint MLP under held-out pair, drug, cell-line, scaffold, and HDAC-class evaluations? Second, do model-derived reversal scores show concordance with experimentally measured LINCS expression responses for compounds kept out of corrected annotation-defined HDAC training? Third, after candidate roles are frozen, are reversal-associated programs coherent under network, pathway, genetic-dependency, and controlled docking analyses? The resulting evidence remains computational and perturbational-transcriptomic; it is intended to select future experiments, not to substitute for them.

2 Methods

2.1 Study design and evidence hierarchy

The revision retained the molecular prediction task and pan-cancer screening framework of the submitted study while replacing unsupported analyses with three frozen computational packages. Package 1 reconstructed the evaluation splits, trained both models across repeated seeds, screened the same compound library, audited structural proximity, quantified candidate-rank stability, and compared predicted with measured LINCS reversal. Package 2 froze candidate roles before generating reversal-associated gene sets, STRING networks, functional enrichment, and DepMap context. Package 3 reframed docking as a controlled sensitivity analysis using crystallographic redocking, an O-methyl zinc-chelation decoy, repeated seeds, and a second HDAC receptor. Four evidence labels were used throughout. “Predicted reversal” denotes a score calculated from a model-generated perturbation vector, whereas “measured transcriptomic reversal” denotes a score calculated from an observed LINCS Level 5 signature matched to official condition metadata. “Biological context” denotes enrichment, network, or genetic-dependency results that are not compound-response assays. “Structural sensitivity” denotes docking results that are not binding measurements. No simulated TCGA survival or stage data, unsupported legacy tumor-mutation-burden conclusions, proxy GDSC validation, or inferred direct targets from reversed genes were included in the revised analysis.

2.2 TCGA pan-cancer disease signatures

Bulk RNA-sequencing cohorts from 22 TCGA solid tumor types were accessed through the Genomic Data Commons [1, 15]: BLCA, BRCA, CESC, CHOL, COAD, ESCA, HNSC, KICH, KIRC, KIRP, LIHC, LUAD, LUSC, PAAD, PCPG, PRAD, READ, SARC, STAD, THCA, THYM, and UCEC. Low-expression genes were filtered, library sizes were normalized by the trimmed mean of M-values, and $\log_2$ counts per million with precision weights were obtained using the *voom* workflow [16]. Tumor-versus-normal differential expression was estimated with empirical-Bayes moderation and Benjamini–Hochberg false-discovery-rate control. The downstream disease vector retained the shared 12,328-gene feature space used by the LINCS response target. Disease signatures were used only for molecular-program reversal; clinical stage, survival, and mutation burden were not modeled. Additional file 2 identifies all 22 GDC projects and records the frozen disease-matrix and observation-mask hashes. Final per-cohort tumor and normal sample counts were not retained in the frozen portable archive and are left blank rather than reconstructed from current GDC holdings.

2.3 LINCS L1000 dataset composition, quality control, and annotation

LINCS L1000 Level 5 treatment signatures were obtained from GSE92742 [8]. Profiles were joined to the Broad signature-metrics table by `sig_id`, retained when the transcriptional activity score was at least 0.2, and excluded when the SMILES field was unavailable, invalid, or marked restricted. Structures were parsed and canonicalized with RDKit [17]. Duplicate signature identifiers were removed. The resulting target matrix contained 55,695 unique Level 5 signatures by 12,328 genes.

Level 5 signatures are consensus expression profiles derived upstream from replicate measurements rather than raw wells. Multiple Level 5 signatures can nevertheless represent the same nominal perturbagen–cell–dose–time combination because different instances, plates, or batches remain. These rows were not described as unique compounds or independent biological replicates. The dataset contained 11,445 valid unique canonical structures, 65 cell lines, 13,314 signature-ID perturbagen tokens, and profiles at 6, 24, and 48 hours (27,226, 28,443, and 26 signatures, respectively). Table 1 reports the full composition. Related signatures were kept together according to the grouping rule of each evaluation split, and candidate measured profiles were aggregated hierarchically as described below.

Table 1: **Composition of the quality-controlled LINCS modeling dataset.** Level 5 signatures are replicate-collapsed LINCS profiles; multiple signatures can remain for the same nominal compound–cell–time–dose condition because of plate or batch structure.

Item	Count	Interpretation
Level 5 signatures	55,695	Modeling units after quality control
LINCS perturbagen tokens	13,314	Signature-ID perturbagen tokens, including instance suffixes
Base perturbagen identifiers	11,527	Normalized BRD identifiers
Unique canonical structures	11,445	Valid structures; no leave-drug train/test overlap
Cell lines	65	Across all signatures
Perturbagen-token-cell combinations	39,266	Descriptive signature-ID combinations
Perturbagen-token-cell-time-dose conditions	52,834	Multiple Level 5 signatures may remain for a condition
Additional signatures beyond unique conditions	2,861	Plate/batch-level Level 5 signatures

The official `siginfo_beta.txt` table was authoritative for candidate validation. We resolved perturbagen identifier, `cmap_name`, perturbation type, official cell label, dose, time, number of contributing measurements, transcriptional activity score, `qc_pass`, and `is_hiq`. All cached signatures matched official signature identifiers. Twelve candidate signatures had cache-versus-official cell-label differences; official labels were used. Target and mechanism-of-action fields were retained from the screening-library annotations when available, but coverage was not assumed complete across all structures. Candidate identities, HDAC annotations, training/test membership, and evidence roles are supplied in Additional file 2.

2.4 Molecular representations and prediction models

Canonical SMILES strings were converted into two molecular representations. The atom-token input was a variable-length matrix in which each atom was represented by a 43-dimensional one-hot element identity, including an “unknown” category. Molecules within a batch were padded and accompanied by a Boolean padding mask. The second representation was a 1,024-bit radius-2 Morgan fingerprint [17, 18].

For atom token i with feature vector $\mathbf{x}_i$, the dual-stream model first applied a learned projection,

$$\mathbf{h}_i^{(0)} = \mathbf{W}_x \mathbf{x}_i + \mathbf{b}_x. \quad (1)$$

Three Transformer encoder layers (hidden dimension 512, eight attention heads, feed-forward dimension 2,048, ReLU activation, dropout 0.1) processed the padded token sequence [20]. Standard Transformer layer normalization was used within each encoder layer. Masked mean pooling and a linear projection produced a 512-dimensional atom representation. Because bond identity and adjacency were not passed to the Transformer, no edge-conditioned message-passing claim is made.

The fingerprint stream used a 1,024-unit linear layer, batch normalization, ReLU, dropout 0.3, and a 512-unit projection. The two 512-dimensional representations were concatenated and passed through a 2,048-unit fusion layer with batch normalization, ReLU, and dropout 0.4 before the 12,328-dimensional output. The strong conventional fingerprint MLP comparator used 1,024- and 2,048-unit hidden layers, batch normalization after both, ReLU, dropout 0.3 and 0.4, and the same output dimension. The dual-stream and fingerprint models contained 38,682,152 and 28,415,016 trainable parameters, respectively.

Crucially, neither model received cell-line identity, dose, time, or other experimental-condition covariates. All rows sharing a canonical structure therefore had the same chemical input even when their measured targets differed by condition. The learned output should be interpreted as a structure-conditioned central tendency across the training conditions, not as a condition-specific response prediction. Accordingly, the leave-cell-line setting tests agreement with profiles from unseen cellular backgrounds under this chemical-only formulation; it does not show that the model learned cell-specific response rules.

Both models minimized mean squared error with AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}), batch size 128, a maximum of 30 epochs, and early stopping after five validation epochs without improvement. The architecture and optimization settings were frozen before the revision benchmark and were not tuned separately by split, seed, candidate, or test performance. We did not conduct or claim an exhaustive hyperparameter search; the purpose of the revision experiments was unbiased evaluation of the submitted model family. Training used automatic mixed precision on an NVIDIA GeForce RTX 4090. Python 3.12.3, PyTorch 2.5.1+cu124, NumPy 2.0.1, pandas 2.3.3, and SciPy 1.13.1 were used. Python, NumPy, PyTorch, and CUDA seeds were fixed; nondeterministic CUDA attention kernels were disabled. Complete settings, selected epochs, runtimes, GPU information, seeds, package versions, and checkpoint hashes are provided in Additional file 1 and the repository manifests.

2.5 Leakage-aware split construction and evaluation

Five evaluation settings were generated from one quality-controlled master table. The drug-cell pair split grouped canonical structure plus cell line. Leave-drug-out grouped canonical structures; leave-cell-line-out grouped cell lines; and the scaffold split grouped Bemis–Murcko scaffolds [19]. Acyclic structures, whose Murcko scaffold is empty, were assigned structure-specific groups rather than one artificial mega-group. The corrected annotation-defined HDAC holdout placed every unambiguously HDAC-annotated structure with matched LINCX modeling profiles in the test set and removed overlap by signature identifier, perturbagen, canonical structure, and drug-cell pair. Thirty of the 39 annotated screening-library structures had matched profiles and contributed 1,856 test signatures. A pre-correction HDAC split was discarded from inferential reporting after its annotation audit found nine omitted profiled structures, including NCH-51 in training; only the corrected test set is used for model conclusions.

The top-level train/test partition used grouping seed 42 and an intended 15% test fraction where group sizes permitted. Within each training partition, 10% was assigned to validation using the same grouping variable and the run seed. The pair split was repeated with seeds 1–5; leave-drug, leave-cell-line, scaffold, and corrected annotation-defined HDAC evaluations used seeds 1–3. All test signature overlaps were zero. The scaffold split additionally had zero scaffold overlap; other splits retained the scaffold overlap expected from their different grouping objectives. Split inventories are reported in Additional file 1.

Performance was summarized by element-wise mean squared error (MSE), mean absolute error (MAE), R^2 , and the mean of profile-wise Pearson and Spearman correlations across the test signatures. Values are reported as mean and sample standard deviation across seeds. Paired model differences used the same split and seed. Correlations were oriented as dual-stream minus fingerprint; errors were oriented as fingerprint minus dual-stream, so positive values always favored the dual-stream model. Two-sided 95% t intervals were calculated across seeds. These small-seed intervals are uncertainty summaries, not evidence of broad population-level significance. Candidate-level strict leave-drug comparisons were reported only for candidates actually assigned to the leave-drug test set.

2.6 Pan-cancer screening and transcriptomic reversal metrics

Each checkpoint screened a frozen library of 28,477 canonical compounds against the 22 TCGA disease signatures. The primary signed weighted transcriptomic reversal score was

$$\text{signed-wTRS}(D, P) = -\frac{\sum_g D_g P_g}{\sum_g |D_g|}, \quad (2)$$

where D_g is the cancer-associated log fold change and P_g is the predicted or measured perturbation for gene g . Larger values indicate stronger signed opposition. The co-primary rank metric was $-\rho_{\text{Spearman}}(D, P)$. A legacy reversal-only weighted-magnitude score was retained in machine-readable sensitivity tables but omitted from primary inference because its corrected annotation-defined HDAC seed stability was weaker.

Signed wTRS is linear in the perturbation vector: multiplication of P by a positive constant multiplies the score by the same constant. It therefore reflects both directional opposition and perturbational amplitude. By contrast, Spearman correlation is invariant to positive rescaling and depends on gene-rank order. Divergence between the two metrics can consequently reflect response magnitude as well as reversal definition, so enrichment under signed wTRS is not generalized to the Spearman metric.

Candidate-rank stability was evaluated across four generalization or distribution-shift regimes (leave drug, leave cell line, scaffold, and corrected annotation-defined HDAC), two models, three seeds, and 22 cancers. The primary analysis used the signed wTRS and Spearman reversal metrics, yielding 48 split-model-seed-metric configurations per candidate. The legacy metric expanded this to 72 configurations only as a sensitivity analysis. Within each configuration, the median rank across cancers was converted to a strength percentile, where 100 denotes the strongest library rank. This order of aggregation prevents 22 cancer rows from being treated as independent experimental replicates and does not imply that every named candidate was unseen under every regime.

Formal library-level HDAC enrichment used a separate cross-generalization consensus. For each metric, the 28,477 compounds were ranked within each of 24 screening runs (four generalization regimes, two models, and three seeds) and each of 22 cancer signatures. Each rank was converted to a within-library rank fraction, where 0 denotes the strongest rank. The 528 run-cancer rank fractions per compound were averaged with equal weight, and compounds were ordered by ascending

mean rank fraction. Enrichment was tested at the fixed revision cutoffs of the top 0.5%, 1%, 5%, and 10%. We evaluated both all annotation-defined HDAC structures (39 of 28,477) and a stricter subset with explicit class I HDAC1/2/3/8 target annotation (26 of 28,477). One-sided Fisher exact tests compared observed with library-expected counts, and Benjamini–Hochberg adjustment was applied across the four cutoffs within each metric and HDAC definition. Signed wTRS and Spearman reversal were tested separately; enrichment under one metric was not treated as evidence for the other. These cutoffs were fixed for the revision analysis but were not prospectively preregistered.

2.7 Official-condition measured LINCS comparison

Candidate profiles were matched to official LINCS metadata. Analyses were performed for all matched profiles, `qc_pass` profiles, and official high-quality (`is_hiq`) profiles; the high-quality stratum was primary. To limit unequal profiling from dominating a candidate, scores were aggregated in four steps: median within candidate–dose–time–official-cell condition, median across conditions within official cell, comparison with each cancer disease vector, and summary across official cell lines and cancer types.

The primary measured set comprised Mocetinostat, NCH-51, TC-H-106, Belinostat, PCI-24781, Panobinostat, Entinostat, and Vorinostat. RG2833 had no measured LINCS signature. Six cached records named Tianeptinaline resolved officially to BG-1010, creating a name–structure conflict; that row was retained only for sensitivity documentation and excluded from primary analyses. Predicted scores from the corrected annotation-defined HDAC ensemble were compared with measured high-quality scores at the candidate–cancer level (eight candidates $\times$ 22 cancers = 176 pairs). Pearson and Spearman 95% intervals used 10,000 crossed bootstrap iterations that independently resampled both candidates and cancers (seed 20260719). Leave-one-candidate-out and leave-one-cancer-out results were calculated as influence diagnostics.

As a secondary, unmatched sensitivity analysis, each candidate’s all-profile measured score was located within the empirical distribution of 11,445 measured canonical compounds for each cancer. Because dose, time, cell background, and quality strata were not matched across this universe, these ranks were not interpreted as a condition-matched non-HDAC null or as independent validation.

Accordingly, the eight-compound comparison was designed to assess predicted–measured transcriptomic concordance only. Without a condition-matched non-HDAC comparator set, it cannot

establish preferential concordance or validation for HDAC compounds relative to alternative perturbagens.

2.8 Candidate identity and structural-neighbor audit

For each named compound and split, exact perturbagen membership, canonical structure, Murcko scaffold, and training/test signatures were enumerated. The maximum Morgan-fingerprint Tanimoto similarity to any training structure, mean similarity among the ten nearest training structures, scaffold exposure, and numbers of neighbors at 0.7, 0.8, and 0.9 were recorded. Exact-structure absence was therefore not treated as sufficient evidence of chemical novelty. Candidate roles were frozen from this audit and measured-profile availability before downstream network, dependency, or docking analyses.

2.9 Frozen reversal-associated genes, networks, and enrichment

For Mocetinostat, NCH-51, and exploratory TC-H-106, reversal-associated genes were defined using directional consensus between corrected annotation-defined HDAC predictions and measured high-quality profiles. A 70% agreement threshold was primary, with 60% and 80% sensitivity analyses; the top 200 genes per candidate were carried forward. Belinostat, Entinostat, and Vorinostat were processed as class-support or reference controls. Candidate tiers were not changed after viewing pathway or network results.

We submitted 11,144 unique measured Entrez identifiers as the custom enrichment background. The g:Profiler service reported an effective statistical domain size of 11,154 after service-side identifier mapping; these two quantities are reported separately rather than treated as interchangeable. Query-gene intersections were reconstructed by aligning each term’s per-query evidence array to the mapped-query order returned in the frozen service response; GO evidence codes were retained in a separate field and were not treated as gene identifiers. All 5,037 term-level intersection counts were verified against the service-reported sizes. Request hashes, cache hashes, service metadata, and the repair audit are supplied with the machine-readable supplement. g:Profiler was used for Gene Ontology biological process, molecular function, and cellular component; KEGG; and Reactome enrichment, with false-discovery-rate control at 0.05 [25]. STRING v12 networks used a required confidence score of 700 and were exported separately for physical and functional associations [26].

Network nodes and enriched terms are described as reversal-associated because they derive from the same predicted/measured expression signal used for ranking; they are not asserted direct drug targets.

2.10 DepMap class I HDAC dependency context

DepMap Public 26Q1 CRISPR gene-effect values for HDAC1, HDAC2, HDAC3, and HDAC8 were joined to official model metadata [32]. All 1,208 models matched metadata and had finite values for the four genes. Pan-cancer medians used 5,000 bootstrap resamples (seed 20260718), and fractions at gene effect ≤ -0.5 and ≤ -1.0 were reported descriptively. For lineages with at least ten models, Kruskal–Wallis tests were followed by Benjamini–Hochberg correction and epsilon-squared effect sizes. Pairwise HDAC relationships used Spearman correlation. Genetic-loss phenotypes were interpreted as target-family context, not compound sensitivity or normal-tissue safety.

2.11 Controlled docking and receptor sensitivity

Docking was redesigned around controls. AutoDock Vina 1.2.7 and Meeko 0.7.1 were used with seeds 1–3, exhaustiveness 32, 20 output modes per run, top-five-pose consensus, and a 2 Å clustering threshold [27, 28]. The primary redocking receptor was the HDAC2–Vorinostat cocrystal 4LXZ because it includes a bound inhibitor and catalytic zinc. The search box was defined by the crystallographic ligand plus 5 Å padding. Standard Vina and the zinc-aware AutoDock4Zn potential were compared [29]. Symmetry-corrected heavy-atom RMSD was evaluated against a prespecified 2 Å redocking gate.

The frozen 4LXZ panel contained Vorinostat as crystallographic positive control; an O-methyl Vorinostat designed to weaken the prespecified zinc-chelating group; Mocetinostat; NCH-51; TC-H-106; Entinostat; and Belinostat. In addition to score, we recorded the nearest ligand heteroatom and prespecified zinc-binding-group distances to catalytic zinc. HDAC1 receptor sensitivity used 4BKX chain B aligned to 4LXZ chain A. Five incomplete side chains remote from the frozen box were conservatively renamed alanine while preserving observed backbone and $C\beta$ coordinates. Sequence identity, $C\alpha$ RMSD, catalytic-zinc alignment, repair distance, seed consensus, and cross-receptor pose differences were recorded before interpretation.

2.12 Statistical analysis, reproducibility, and reporting

Unless stated otherwise, tests were two-sided and descriptive. Benjamini–Hochberg correction was used within each stated enrichment or lineage-analysis family. No candidate was excluded because of an unfavorable downstream value; identity eligibility and candidate tiers were fixed first. The repository distinguishes original analysis outputs, analyses regenerated during revision, retained derived evidence, large inputs or run artifacts that are not redistributed, and intermediates that are no longer available. Versioned derived tables, figures, split inventories, source identifiers, seeds, selected hashes, software versions, runtimes, and audit logs support direct verification of the reported summaries and recompilation of the documents from frozen table and figure assets. The split, training, screening, and downstream analysis code is archived, but a self-contained bitwise end-to-end rerun from the GitHub repository and current public inputs alone is not guaranteed because large GPU checkpoints and prediction arrays and specified analysis-time intermediates were not retained in the portable archive. Stage-specific requirements and limitations are documented in the repository reproducibility statement. Additional file 1 provides human-readable supplementary methods and tables, and Additional file 2 provides machine-readable tables; large public source datasets remain at their originating repositories.

3 Results

3.1 Dataset and split audits distinguish signatures, structures, and conditions

The 55,695 modeling rows represented LINCS Level 5 signatures rather than 55,695 unique compounds or drug–cell pairs. They comprised 11,445 unique valid canonical structures and 65 cell lines (Table 1). Signature identifiers contained 13,314 perturbagen-instance tokens and 52,834 distinct perturbagen-token–cell–time–dose combinations; 2,861 additional signatures reflected repeated Level 5 entries for nominal conditions. The three recorded times were strongly concentrated at 6 and 24 hours, with only 26 profiles at 48 hours. These distinctions are important because profile count alone overstates the number of independent chemical entities.

The split inventory confirmed zero signature-identifier overlap in every setting. The pair split contained 47,376 training and 8,319 test signatures, leave drug 47,728 and 7,967, leave cell line 50,520

and 5,175, scaffold 47,851 and 7,844, and corrected annotation-defined HDAC 53,839 and 1,856. The corresponding train/test unique-structure counts were 10,582/3,827, 9,728/1,717, 11,170/2,826, 9,831/1,614, and 11,415/30. Scaffold overlap was 2,489, 686, 1,914, 0, and 11, respectively, consistent with the different grouping objectives. Thus, only the scaffold split tested scaffold-level separation, while the HDAC holdout tested unseen exact structures within a biologically defined class. Relative to the invalid pre-correction split, the corrected holdout added 291 profiles and nine structures and removed NCH-51 from training; the complete 30-structure inventory is reported in Additional files 1 and 2.

3.2 Pan-cancer screening shows signed-wTRS-specific class I HDAC enrichment

The corrected annotation-defined HDAC ensemble produced a coherent but heterogeneous pan-cancer landscape for the frozen candidate set (Figure 1, panels A–B). The separate 24-run, 22-cancer cross-generalization consensus used for formal library-level testing showed strong enrichment of the explicitly annotated class I HDAC subset under signed wTRS (panel C). At the fixed top 0.5%, 1%, 5%, and 10% revision cutoffs, 4, 5, 11, and 12 of 26 class I structures were observed, versus 0.13, 0.26, 1.30, and 2.60 expected from a 28,477-compound library. The corresponding fold enrichments were 30.6, 19.2, 8.46, and 4.62; one-sided Fisher P values ranged from 1.81×10^{-8} to 8.36×10^{-6} and all within-family FDR values were below 10^{-4} . The co-primary Spearman reversal score did not significantly enrich the same subset at these cutoffs. Because signed wTRS is proportional to perturbational amplitude while Spearman correlation is scale invariant, the divergence may partly reflect overall response magnitude. The evidence therefore supports enrichment only under signed wTRS, not general enrichment across reversal definitions. Results for the broader 39-structure annotation-defined HDAC set and the legacy score are provided as fixed companion and sensitivity analyses in Additional file 2.

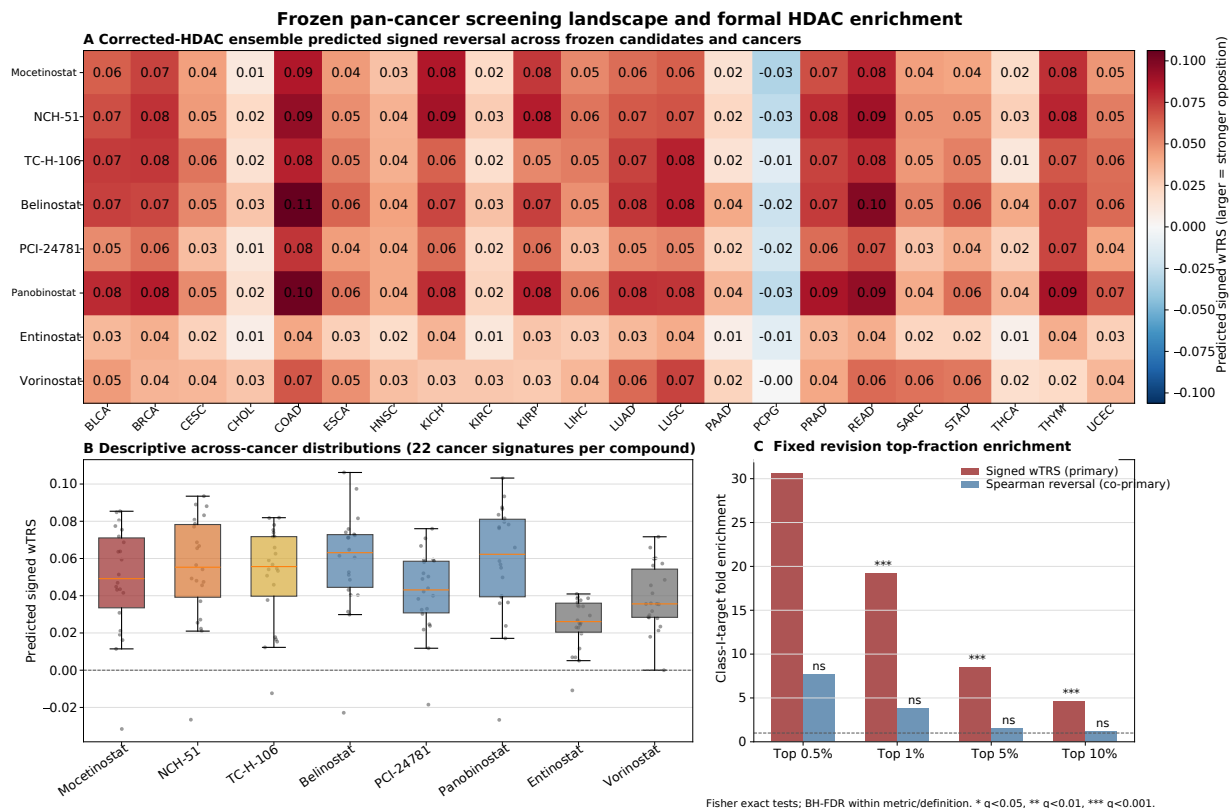


Figure 1: Corrected annotation-defined HDAC pan-cancer screening landscape and formal enrichment. (A) Corrected-HDAC-ensemble predicted signed wTRS for eight interpretable candidates or controls across 22 TCGA disease signatures. (B) Across-cancer score distributions from the same corrected-HDAC ensemble show candidate heterogeneity rather than uniform pan-cancer activity. (C) Observed-to-expected enrichment of the explicitly annotated class I HDAC subset at fixed revision cutoffs, using the separate consensus that equally averaged 528 rank-fraction observations per compound (24 generalization runs $\times$ 22 cancers; 0 denotes the strongest rank). Signed wTRS was significantly enriched after FDR control; Spearman reversal was not. These are computational prioritization scores, not efficacy measurements.

3.3 Repeated generalization benchmarks show no measurable dual-stream gain

Across the five settings, differences between the dual-stream model and the strong conventional fingerprint MLP comparator were small relative to seed variability and changed orientation by metric (Table 2). In the five-seed drug-cell pair split, the dual-stream model achieved MSE 0.882, MAE 0.704, Pearson $r = 0.259$, and Spearman $\rho = 0.237$; the fingerprint MLP achieved 0.881, 0.703, 0.263, and 0.241. The mean paired dual-stream Pearson difference was -0.0034 (95% CI -0.0170 to 0.0102). Its Spearman, MSE, and MAE intervals also crossed zero.

Table 2: **Leakage-aware generalization across five frozen evaluation settings.** Values are mean (seed SD); the pair split used five seeds and all other settings used three. Correlations are means of profile-wise coefficients. The HDAC row uses the corrected annotation-defined holdout only; its 1,856 test profiles cover all 30 held-out HDAC structures, whereas Table 3 reports candidate-level subsets. No split establishes consistent dual-stream superiority.

Split	Model	Test n	MSE $\downarrow$	MAE $\downarrow$	R^2 $\uparrow$	Pearson r $\uparrow$	Spearman ρ $\uparrow$
Drug-cell pair	Dual stream	8,319	0.882 (0.004)	0.704 (0.003)	0.108 (0.004)	0.259 (0.009)	0.237 (0.009)
Drug-cell pair	Fingerprint MLP	8,319	0.881 (0.001)	0.703 (0.001)	0.110 (0.002)	0.263 (0.005)	0.241 (0.005)
Leave drug out	Dual stream	7,967	0.919 (0.003)	0.717 (0.001)	0.052 (0.003)	0.188 (0.000)	0.166 (0.001)
Leave drug out	Fingerprint MLP	7,967	0.919 (0.001)	0.716 (0.001)	0.052 (0.001)	0.191 (0.003)	0.170 (0.003)
Leave cell line out	Dual stream	5,175	0.942 (0.004)	0.714 (0.000)	0.121 (0.003)	0.266 (0.010)	0.243 (0.010)
Leave cell line out	Fingerprint MLP	5,175	0.941 (0.007)	0.713 (0.002)	0.122 (0.006)	0.272 (0.010)	0.249 (0.010)
Scaffold split	Dual stream	7,844	0.973 (0.000)	0.738 (0.002)	0.062 (0.000)	0.185 (0.006)	0.163 (0.007)
Scaffold split	Fingerprint MLP	7,844	0.971 (0.002)	0.736 (0.001)	0.064 (0.002)	0.186 (0.001)	0.165 (0.002)
Corrected annotation-defined HDAC	Dual stream	1,856	1.326 (0.026)	0.839 (0.005)	0.131 (0.017)	0.378 (0.009)	0.345 (0.010)
Corrected annotation-defined HDAC	Fingerprint MLP	1,856	1.333 (0.015)	0.841 (0.004)	0.127 (0.010)	0.379 (0.007)	0.344 (0.006)

The same pattern held under stricter distribution shifts. Mean Pearson correlations for dual stream versus fingerprint were 0.188 versus 0.191 in leave drug, 0.266 versus 0.272 in leave cell line, and 0.185 versus 0.186 in scaffold split. In the corrected annotation-defined HDAC holdout, dual-stream versus fingerprint values were MSE 1.326 versus 1.333, MAE 0.839 versus 0.841, R^2 0.131 versus 0.127, Pearson 0.378 versus 0.379, and Spearman 0.345 versus 0.344. Here, average errors and Spearman correlation marginally favored the dual stream, whereas Pearson correlation marginally favored the fingerprint MLP. All five paired 95% intervals, including R^2 , crossed zero. The evidence therefore shows no measurable performance gain from the dual-stream architecture over the strong conventional comparator.

Training time reflected the larger dual-stream architecture. Depending on split, mean end-to-end training and evaluation time was 152–303 seconds per dual-stream run and 46–76 seconds per fingerprint run on the RTX 4090. Early stopping selected mean epochs between 3.0 and 10.0 for the dual stream and 3.7 and 11.7 for the fingerprint model across settings. Thus, the added parameter and runtime cost was not accompanied by a measurable predictive gain. These observed costs and every run-level value are reported in Additional file 1.

3.4 Strict candidate-level generalization is heterogeneous

All eight unambiguous measured HDAC candidates or controls were absent from corrected annotation-defined HDAC training by exact structure. Candidate-level profile prediction nevertheless varied. For the dual-stream model, mean Pearson correlations ranged from 0.233 for PCI-24781 to 0.548 for

Belinostat. The fingerprint range was 0.241 for Entinostat to 0.605 for Belinostat. The dual stream had higher mean correlation for Entinostat, TC-H-106, and Vorinostat, whereas the fingerprint model was higher for Belinostat, Mocetinostat, NCH-51, PCI-24781, and Panobinostat. This heterogeneity is inconsistent with selecting one architecture as universally better.

Strict leave-drug candidate evaluation was possible only for Mocetinostat (14 signatures across 12 cell lines) and PCI-24781 (19 signatures across 13 cell lines). The matching profile counts in the corrected annotation-defined HDAC holdout are expected because both splits assigned every eligible signature for each of these two canonical structures to the test set. The overall partitions are distinct: leave drug contained 7,967 profiles across 1,717 test structures, whereas the corrected HDAC holdout contained 1,856 profiles across 30 annotated structures. For Mocetinostat, mean Pearson correlations were 0.483 (dual stream) and 0.504 (fingerprint), with a paired dual-stream difference of -0.020 (95% CI -0.140 to 0.099). For PCI-24781, the means were 0.308 and 0.540, and the paired difference was -0.232 (95% CI -0.450 to -0.015). No strict candidate comparison supported a dual-stream advantage.

3.5 Official high-quality LINCS profiles provide measured transcriptomic support

Official metadata identified 1,212 measured profiles across nine named rows before identity exclusions. The eight-compound primary set contained 12 high-quality Belinostat profiles, 67 Entinostat, 8 Mocetinostat, 8 NCH-51, 13 PCI-24781, 84 Panobinostat, 4 TC-H-106, and 487 Vorinostat. Official high-quality profiles covered 4–35 cell lines per candidate after condition-first aggregation. Mocetinostat, NCH-51, TC-H-106, and PCI-24781 were measured only at $10\text{ }\mu\text{M}$; the larger reference sets spanned multiple doses. Most profiles were measured at 6 or 24 hours; Vorinostat additionally included 48 hours. These imbalances motivated the prespecified hierarchical median rather than pooling all signatures equally.

All eight primary compounds had positive median signed reversal across the 22 cancer disease signatures (Figure 2). Median signed wTRS values were 0.262 for Mocetinostat, 0.194 for NCH-51, 0.119 for TC-H-106, 0.140 for Belinostat, 0.139 for PCI-24781, 0.191 for Panobinostat, 0.082 for Entinostat, and 0.054 for Vorinostat. Positive signed reversal occurred in 95.5% of cancers for all except TC-H-106 (90.9%). Rank-based measured reversal gave the same directional conclusion,

with median values from 0.043 for Vorinostat to 0.137 for Mocetinostat. Quality-pass and all-profile analyses are provided as sensitivity strata in Additional file 2.

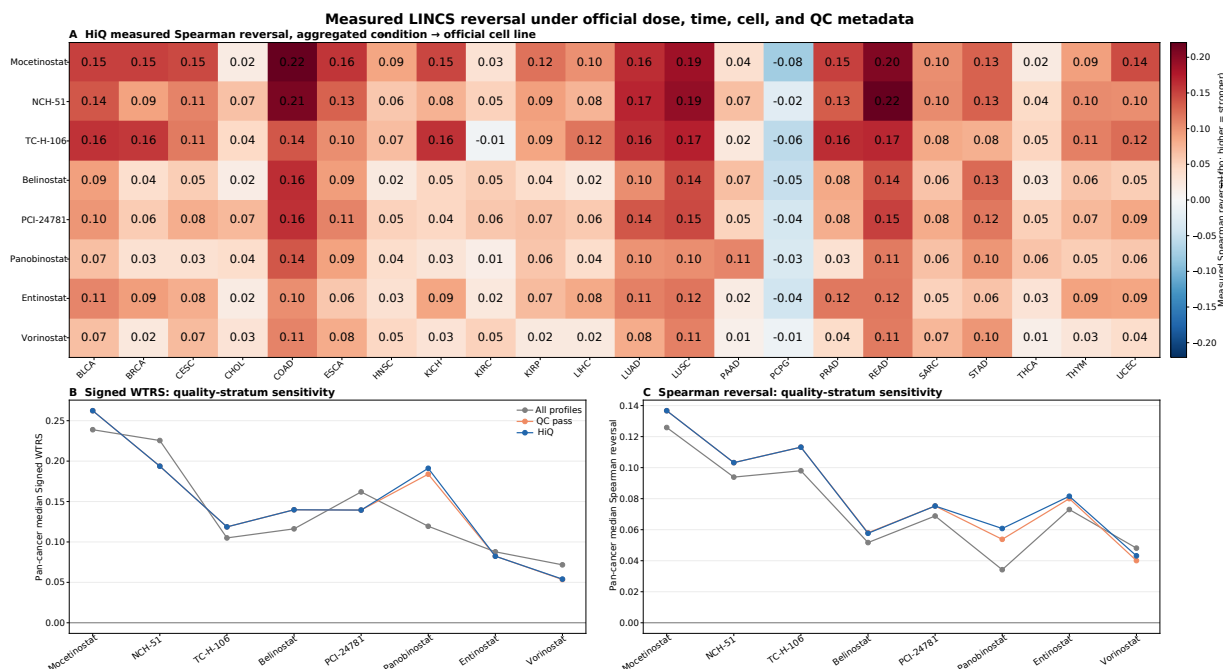


Figure 2: **Measured transcriptomic reversal under official LINC conditions.** (A) Candidate-by-cancer heatmap of official high-quality measured Spearman reversal after condition-first and official-cell-first aggregation. (B) Pan-cancer median signed wTRS and (C) pan-cancer median Spearman reversal across all-profile, QC-pass, and HiQ quality strata. Connected points show quality-stratum sensitivity; they are not boxes or independent replicates. Measured reversal is experimental expression evidence, not direct evidence of viability, efficacy, or binding. RG2833 had no measured profile, and Tianeptinaline/BG-1010 was excluded because its name and official identity conflicted.

3.6 Predicted and measured LINC reversal are concordant for both encoders

Across 176 candidate–cancer pairs, corrected annotation-defined HDAC ensemble predictions were positively associated with official high-quality measured reversal (Figure 3). For signed wTRS, the dual-stream model had Pearson $r = 0.642$ (crossed-bootstrap 95% CI 0.365–0.838) and Spearman $\rho = 0.640$ (0.297–0.840), whereas the fingerprint MLP had $r = 0.805$ (0.601–0.911) and $\rho = 0.803$ (0.554–0.919). For rank-based reversal, the dual-stream values were $r = 0.813$ (0.634–0.919) and $\rho = 0.801$ (0.594–0.909); fingerprint values were $r = 0.853$ (0.710–0.921) and $\rho = 0.830$ (0.653–0.910). The crossed bootstrap resampled both eight candidates and 22 cancers, so it does not treat the 176 pairs as independent. Every interval remained above zero, and leave-one-candidate-out and leave-

one-cancer-out diagnostics retained positive coefficients. These results establish predicted–measured transcriptomic concordance for the eight evaluated compounds and provide no evidence that the dual-stream architecture is superior. Because no condition-matched non-HDAC comparator set was available, they do not establish preferential concordance or validation relative to alternative perturbagens. The analysis also remains internal to held-out compounds from the LINCS resource rather than constituting validation on an independent experimental platform.

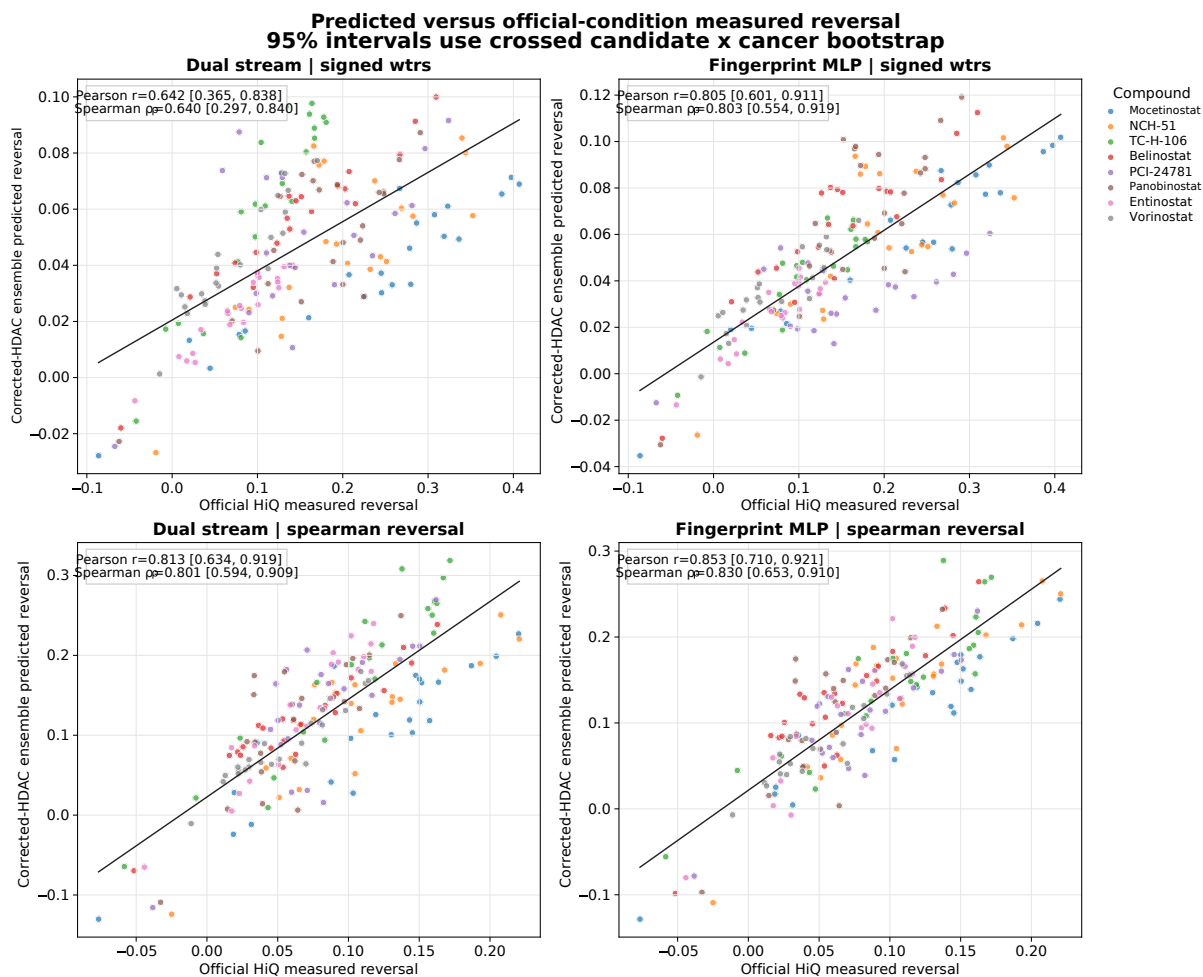


Figure 3: **Corrected annotation-defined HDAC predictions versus measured high-quality LINCS reversal.** Each point is one of 176 hierarchically aggregated candidate–cancer pairs (eight candidates $\times$ 22 cancers), not a raw LINCS signature; Table 3 reports the underlying candidate-specific corrected-HDAC and high-quality profile counts. Solid black lines are ordinary-least-squares visual summaries and are not model calibration curves. Pearson and Spearman coefficients and their 95% intervals from 10,000 crossed bootstrap iterations are printed in the panels and reported in the text. Point-level error bars are not shown because uncertainty was estimated by independently resampling the crossed candidate and cancer clusters. The positive associations support predicted–measured transcriptomic concordance within the eight held-out LINCS compounds; without a condition-matched non-HDAC comparator, they do not establish preferential validation or therapeutic activity.

3.7 Identity, structural exposure, and stability determine candidate tiers

Candidate selection was revised from a three-name narrative to an evidence-tiered hierarchy (Table 3). Mocetinostat had no exact structure or scaffold in corrected annotation-defined HDAC training, a maximum training-neighbor Tanimoto similarity of 0.537, strict leave-drug evaluation, eight

high-quality measured profiles, and a primary 48-configuration median strength percentile of 92.2 (100 = strongest). It was therefore retained as the core computational hypothesis. NCH-51 had eight high-quality profiles and a primary strength percentile of 88.5, but its Murcko scaffold occurred in corrected annotation-defined HDAC training; it was retained as secondary with this caveat.

Table 3: Frozen candidate evidence tiers. Corrected-HDAC profile n is the number of quality-controlled modeling signatures assigned to the candidate in the corrected annotation-defined HDAC test set; HiQ n is the number of official LINCS high-quality profiles. Stability is the median strength percentile (100 = strongest) across the 48 primary split-model-seed-metric configurations using signed wTRS and Spearman reversal; the 72-configuration legacy-inclusive sensitivity analysis is reported in Additional file 2. Identity and exposure rules override descriptive rank.

Candidate	Frozen role	Corrected-HDAC profile n	Max train similarity	HiQ n	Stability percentile	Evidence interpretation
Mocetinostat	core candidate	14	0.537	8	92.2	Primary computational hypothesis
NCH-51	secondary candidate	17	0.431	8	88.5	Measured support; scaffold-exposure caveat
TC-H-106	original method sensitive	6	0.789	4	84.3	Exploratory; limited HiQ coverage and close neighbor
Belinostat	class support	24	0.500	12	70.0	HDAC-class / measured transcriptomic support
PCI-24781	class support	19	0.410	13	57.6	HDAC-class / measured transcriptomic support
Panobinostat	class support	126	0.381	84	60.6	HDAC-class / measured transcriptomic support
Entinostat	reference control	101	0.558	67	77.9	Established HDAC reference control
Vorinostat	reference control	899	0.697	487	69.5	Established HDAC reference control
RG2833	prediction only	0	0.574	0	72.0	Prediction only; no measured LINCS profile
Tianeptinaline/BG-1010	identity conflict excluded	0	1.000	1	90.0	Identity conflict; excluded despite high descriptive rank

TC-H-106 had four high-quality profiles, a primary strength percentile of 84.3, and a closest corrected annotation-defined HDAC training neighbor of 0.789. It was downgraded to exploratory/original-method-sensitive rather than selected as the principal candidate. RG2833 had neither a training nor test signature and no measured LINCS profile; it remains prediction-only. Tianeptinaline/BG-1010 had exact training exposure and an unresolved name-structure conflict and is excluded, even though its descriptive rank was high. Established HDAC inhibitors were retained as class-support or reference compounds rather than promoted as new discoveries.

Rank robustness was substantial for some candidates but variable across analytical choices (Figure 4). In the 24 primary matched split-seed-metric comparisons, dual-stream versus fingerprint rank agreement across all resolved candidates had mean Spearman $\rho = 0.756$, median 0.830, and minimum 0.115. Within the eight eligible measured compounds, the corresponding values were 0.718, 0.833, and 0.119. Thus, high median percentiles do not imply uniform stability under every split and metric; legacy-inclusive comparisons are reported separately in Additional file 2.

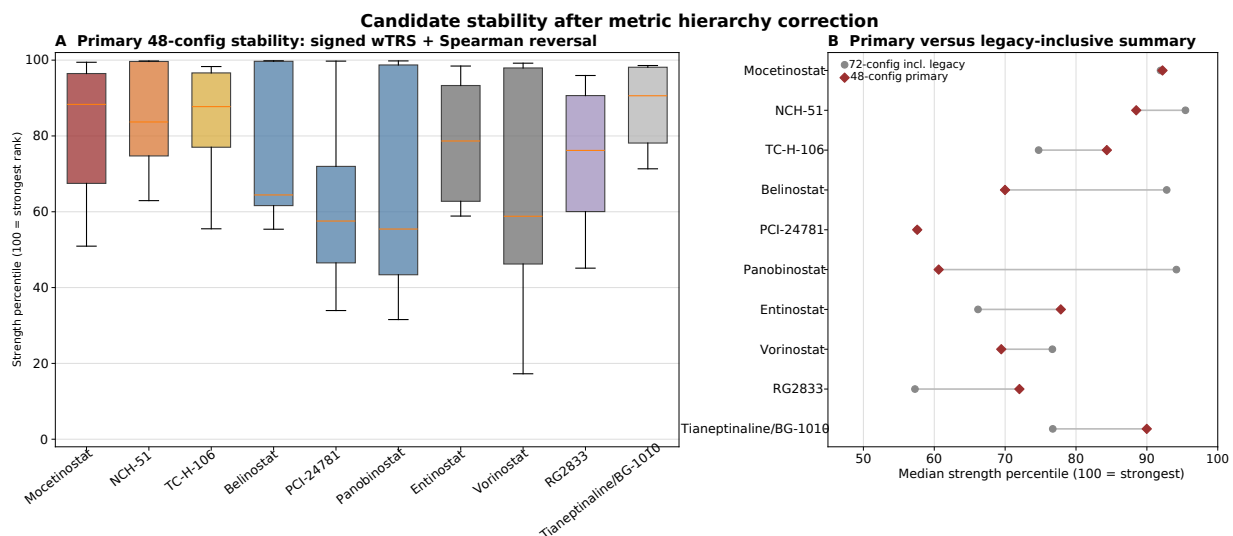


Figure 4: Candidate stability across split, model, seed, cancer, and reversal definition. Candidate-level strength percentiles (100 = strongest) were calculated within the 28,477-compound library after first taking the median across 22 cancers for each run configuration. The primary analysis summarizes 48 configurations using signed wTRS and Spearman reversal; the 72-configuration legacy-inclusive result is shown only as sensitivity analysis. Roles were frozen using identity, measured-profile availability, strict split membership, and structural proximity before downstream biology. RG2833 is prediction-only; Tianeptinaline/BG-1010 is an excluded sensitivity row.

The structural audit further explains why pair-level performance cannot be interpreted as unseen-drug generalization (Figure 5). In the pair split, exact candidate structures were present in training for every measured candidate with a test profile. In corrected annotation-defined HDAC training, exact structures were absent for all eligible HDAC test compounds, but scaffolds remained for NCH-51, Belinostat, Panobinostat, and Vorinostat. The maximum similarities ranged from 0.381 for Panobinostat to 0.789 for TC-H-106. Candidate claims were therefore calibrated to both exact and near-neighbor exposure.

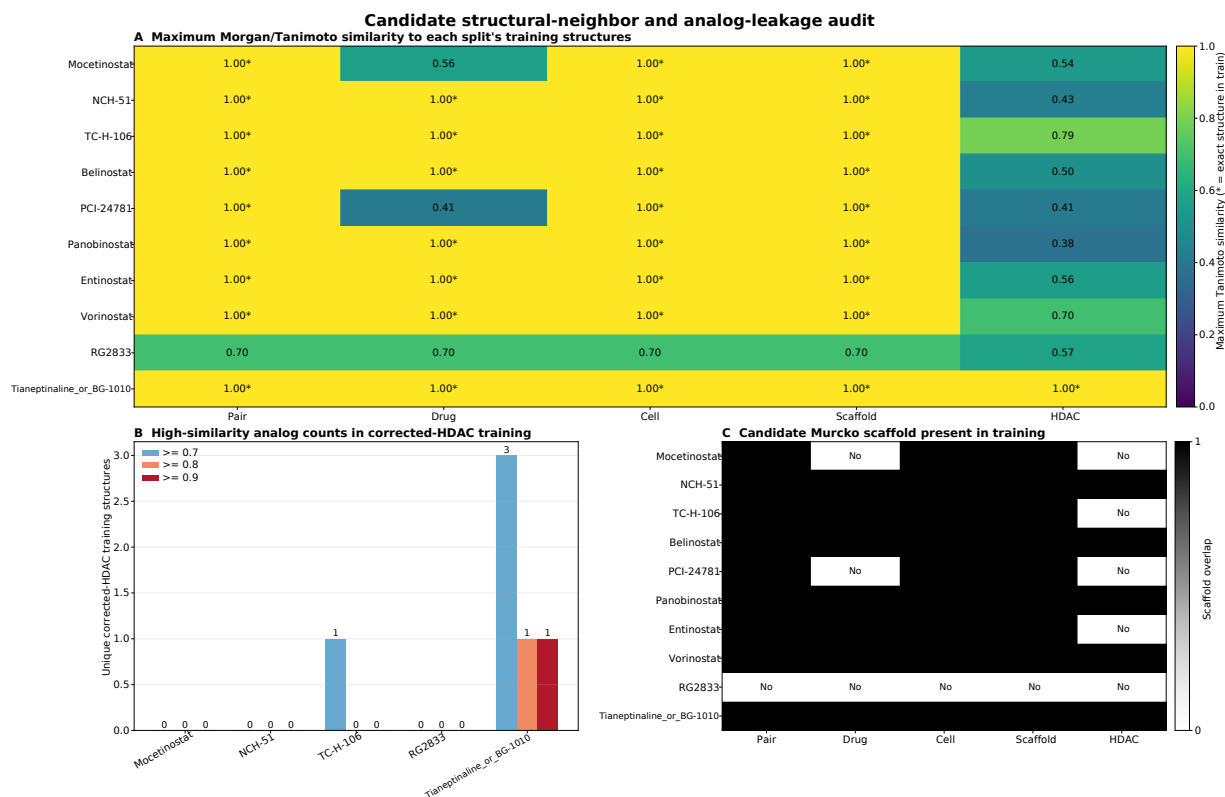


Figure 5: **Training-set exposure and structural-neighbor audit.** Panels report split-level train/test structure and scaffold overlap, candidate membership, and nearest-training-structure Tanimoto similarity. Exact pair separation does not imply unseen-compound evaluation. The scaffold split alone has zero scaffold overlap; the corrected annotation-defined HDAC split removes exact HDAC structures while documenting residual scaffold or analog proximity.

3.8 Reversal-associated networks converge on cell-cycle programs but are not direct targets

At the fixed 70% revision consensus threshold, 200 reversal-associated genes were selected for each of the three prioritized candidates and three HDAC controls, and all mapped to STRING. Functional networks contained 577 edges for Mocetinostat, 944 for NCH-51, and 243 for TC-H-106, with largest components of 66, 86, and 65 nodes, respectively. Physical networks were sparser, containing 52, 59, and 52 edges. These differences show that interaction database type materially changes apparent network density.

Mocetinostat and NCH-51 were enriched for mitotic cell-cycle and chromosome-segregation programs. Representative terms included “mitotic cell cycle process” for Mocetinostat (GO biological process FDR 7.66×10^{-30}), “chromosome segregation” for NCH-51 (8.04×10^{-19}), and Reactome

“Cell Cycle, Mitotic” for both (4.71×10^{-25} and 2.13×10^{-30} , respectively). TC-H-106 showed fewer significant terms; its leading biological process was ribonucleoprotein-complex biogenesis (FDR 3.48×10^{-12}), while Reactome DNA replication (FDR 2.53×10^{-6}) and KEGG spliceosome (FDR 1.60×10^{-4}) were also significant. This profile is more heterogeneous and does not support the original presentation of a uniformly dominant TC-H-106 cell-cycle mechanism.

Gene-set overlap with references was partial rather than complete. Jaccard similarities were 0.274 for Mocetinostat–Entinostat, 0.187 for Mocetinostat–Vorinostat, 0.227 for NCH-51–Entinostat, 0.333 for NCH-51–Vorinostat, 0.325 for TC-H-106–Entinostat, and 0.183 for TC-H-106–Vorinostat. These results support a related HDAC-class transcriptional context with candidate-specific components. Because the selected genes were derived from the same reversal signal used for ranking, enrichment and network topology are complementary annotation rather than independent mechanism validation.

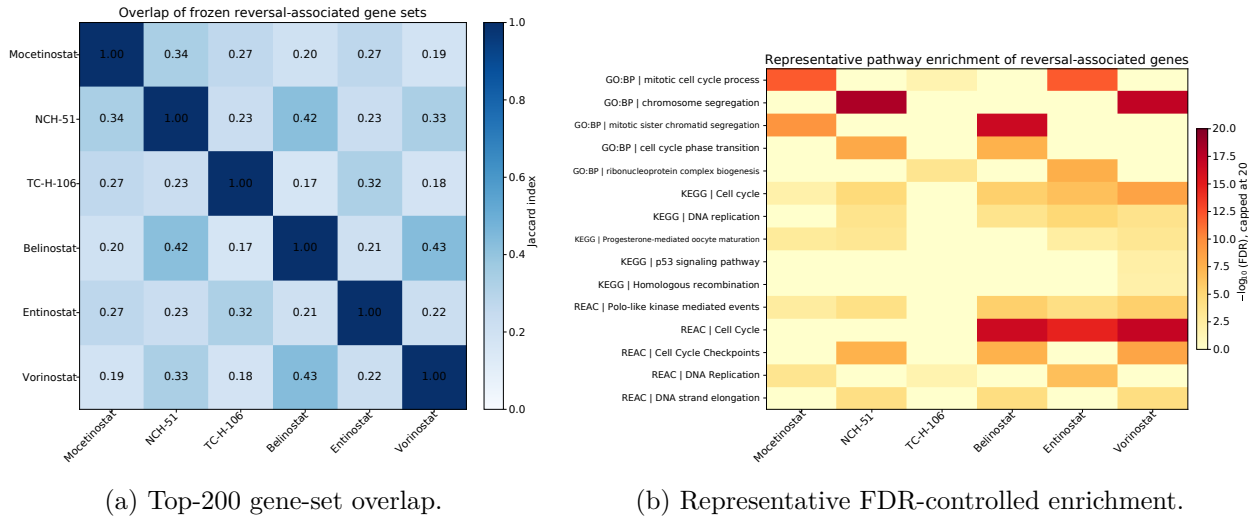


Figure 6: Reversal-associated gene-set overlap and pathway annotation after candidate freezing. (A) Jaccard overlap among candidate and reference top-200 gene sets. (B) Representative GO, KEGG, and Reactome terms after gene-set redundancy filtering, using 11,144 submitted measured Entrez identifiers as the custom background; g:Profiler reported an effective mapped domain size of 11,154. Term intersections were reconstructed from the frozen mapped-query order, and GO evidence codes were stored separately from Entrez gene identifiers. These analyses annotate the reversal signal and are not independent mechanism validation. Full gene sets, term statistics, FDR values, and the intersection audit are supplied in Additional file 2.

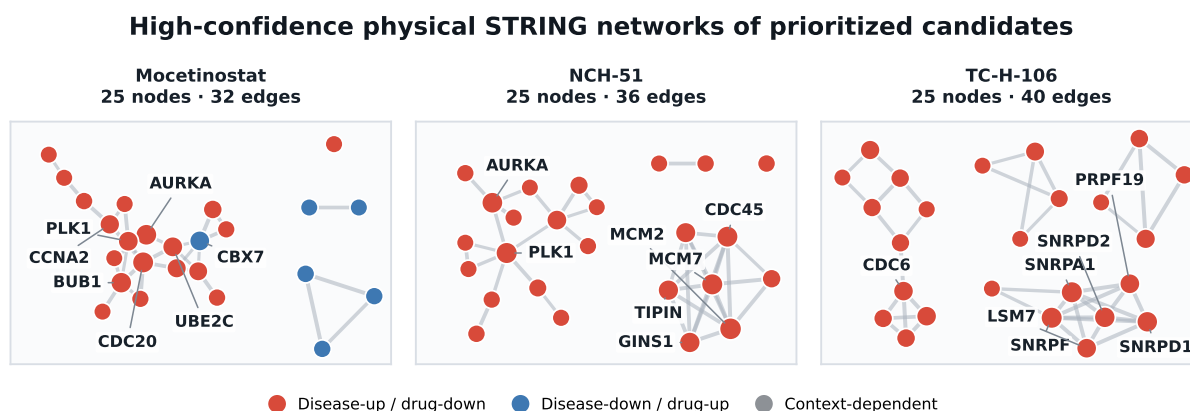


Figure 7: **High-confidence physical STRING networks for the three prioritized candidates.** Networks use STRING v12 required score ≥ 700 . For legibility, each panel displays the 25 nodes with greatest weighted degree and labels the seven leading hubs; complete 200-gene node lists, physical and functional edges, and candidate/control networks are supplied in Additional files 1 and 2. Nodes are reversal-associated genes and must not be interpreted as direct drug targets.

3.9 DepMap identifies HDAC3 as the dominant pan-cancer genetic dependency

Across 1,208 DepMap models, median gene effects were -0.107 for HDAC1 (95% bootstrap CI -0.118 to -0.093), 0.001 for HDAC2 (-0.008 to 0.013), -1.215 for HDAC3 (-1.249 to -1.188), and -0.121 for HDAC8 (-0.134 to -0.107). The fractions at or below -0.5 were 4.47%, 3.15%, 95.12%, and 4.88%, respectively; 68.21% of models had HDAC3 effect at or below -1.0 . Thus, the broad dependency result is driven by HDAC3 rather than HDAC1. All four genes showed FDR-significant lineage heterogeneity across 24 eligible lineages, but epsilon-squared values were modest (0.042–0.083). Pairwise relationships were weak: the largest absolute Spearman correlation was 0.151 between HDAC1 and HDAC2. DepMap therefore supplies family-level genetic-loss context, not evidence that a named inhibitor will reproduce knockout effects or show uniform anti-cancer activity.

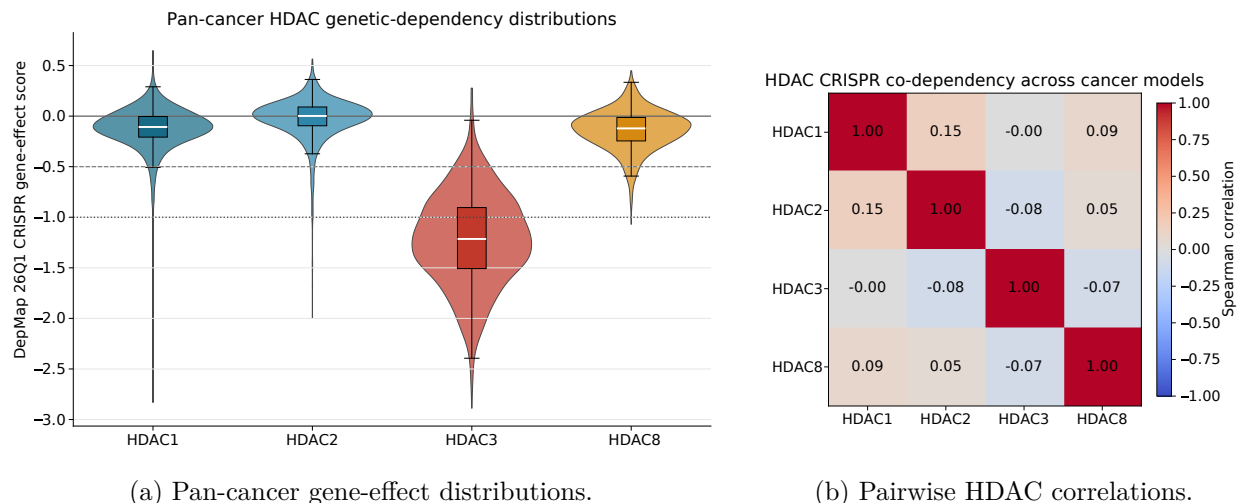


Figure 8: **DepMap Public 26Q1 class I HDAC dependency context.** (A) Overall CRISPR gene-effect distributions across 1,208 models; more negative values indicate greater loss-of-function effect. (B) Weak pairwise Spearman relationships among HDAC1, HDAC2, HDAC3, and HDAC8. HDAC3 is the dominant pan-cancer dependency; these genetic data are not compound-response or safety measurements. The complete heatmap of eligible-lineage medians is provided in Additional file 1.

3.10 Controlled docking demonstrates score and receptor sensitivity

Standard Vina failed to reproduce the crystallographic Vorinostat pose in HDAC2 4LXZ: its selected consensus-medoid RMSD was 4.771 Å. AutoDock4Zn passed the prespecified gate with selected-medoid RMSD 1.091 Å and a crystallographic zinc-binding-group distance of 2.12–3.11 Å. This control established that a zinc-aware potential was necessary for interpreting the metalloprotein pocket in this protocol.

All seven 4LXZ panel compounds produced pose clusters represented across all three seeds, but score did not guarantee the intended chemistry. The O-methyl Vorinostat decoy had a more favorable median score than Vorinostat (−7.998 versus −7.491 kcal/mol) while placing its prespecified zinc-binding group 6.80–8.99 Å from zinc, compared with 2.12–3.11 Å for Vorinostat. Mocetinostat and Entinostat produced favorable scores (−11.024 and −10.703 kcal/mol) but their prespecified zinc-binding-group atoms were more than 11 Å from zinc in the selected clusters, even though another heteroatom approached zinc. No compound was removed post hoc on this basis; the result instead limits score-only interpretation.

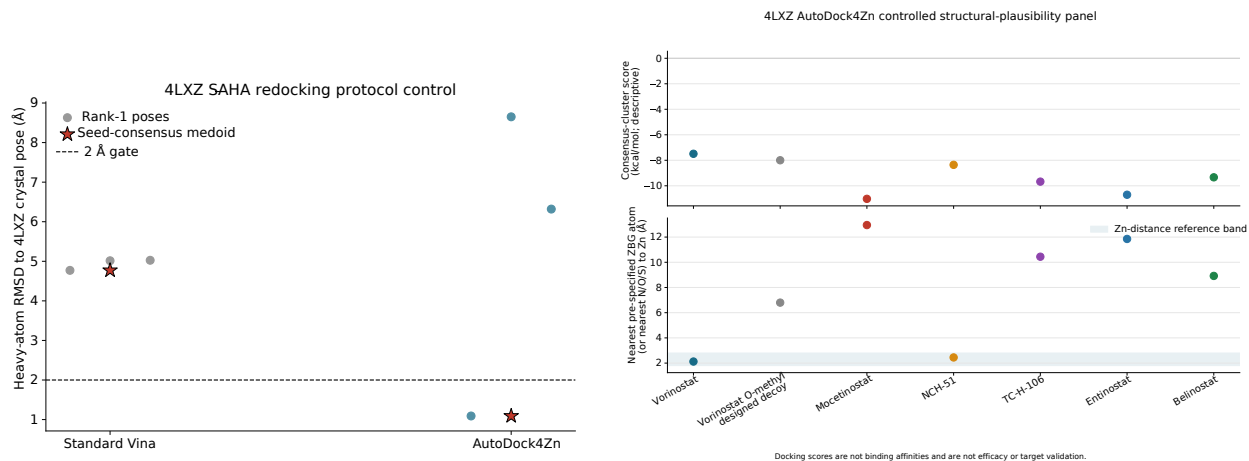
HDAC1 4BKX aligned well to HDAC2 4LXZ (93.7% aligned sequence identity, C α RMSD

0.507 Å, catalytic-zinc distance 0.121 Å). Nevertheless, selected cross-receptor pose differences were compound dependent: fixed-frame medoid RMSD ranged from 1.080 Å for Mocetinostat to 8.787 Å for Belinostat, and score changes ranged from -0.620 to 1.421 kcal/mol. Docking therefore documents protocol and receptor sensitivity together with unresolved pose plausibility; it does not prove HDAC engagement, isoform selectivity, or efficacy.

4 Discussion

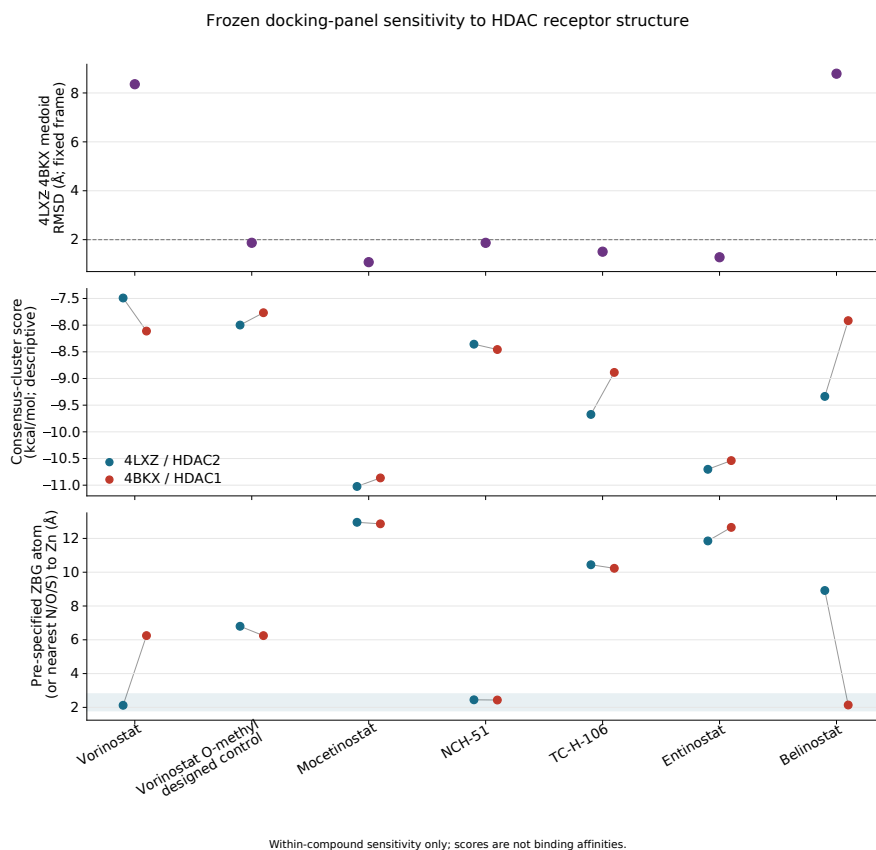
The principal model result is the absence of a measurable dual-stream advantage. The atom-token/fingerprint architecture did not consistently outperform the strong conventional fingerprint MLP comparator trained on the same expression targets and splits. Across pair, leave-drug, leave-cell-line, and scaffold settings, the fingerprint model was slightly better on average for all five reported metrics. In the corrected annotation-defined HDAC holdout, the dual stream had marginally lower errors, higher R^2 , and marginally higher Spearman correlation, while the fingerprint model had marginally higher Pearson correlation. Seed-paired intervals did not establish an advantage, and the dual stream required more parameters and longer observed training times. Its additional representation complexity is therefore not justified by the current benchmarks. This result is consistent with broader evidence that learned molecular representations do not uniformly outperform fixed representations and that dataset properties, chemical-space coverage, splitting, and evaluation choices can dominate apparent performance [37].

This calibration shifts the manuscript’s center of gravity from architectural novelty to evaluation reliability. The principal contribution is an auditable end-to-end framework: data curation, entity-aware generalization, a corrected class holdout, same-data model comparison, structural-proximity and applicability assessment, official-condition measured concordance, frozen candidate roles, and separate biological and structural evidence layers. Related ligand-based workflows have likewise combined curated inputs with applicability-domain assessment and external or temporal validation, illustrating why model utility cannot be inferred from architectural complexity alone [38, 39]. The current work does not claim that concatenating familiar components constitutes a new state of the art, nor does it relabel published architectures as proxies and assign incomparable numbers. A future bond-aware residual graph encoder may improve the atom branch, but changing the primary



(a) Standard Vina versus AutoDock4Zn redocking.

(b) Frozen 4LXZ control/candidate panel.



(c) 4LXZ/4BKX receptor sensitivity.

Figure 9: Controlled zinc-aware docking and receptor sensitivity. (A) Crystallographic Vorinostat redocking passed only with the zinc-aware potential. (B) The positive control, O-methyl zinc-chelation decoy, candidates, and reference HDAC inhibitors were compared using score, seed consensus, and catalytic-zinc geometry. (C) Repeating the frozen panel in aligned HDAC1 4BKX showed compound-dependent pose changes. Docking is reported as structural sensitivity, not binding validation.

model during this revision would confound the completed audit and is unnecessary for the present framework-level conclusion.

The predicted-measured LINCS comparison is the strongest candidate-specific transcriptomic evidence in this study. Predictions from both encoders were concordant with high-quality measured reversal for eight compounds held out from corrected annotation-defined HDAC training, and crossed candidate-cancer intervals plus leave-one-out diagnostics showed that this was not attributable to treating all 176 pairs as independent. Because no condition-matched non-HDAC comparator set was available, the result does not establish preferential concordance for the HDAC set relative to alternative perturbagens. Experimentally measured perturbational expression closes part of the gap between generated signatures and observed response, but not the gap to anti-cancer efficacy. LINCS cell-line transcription at particular doses and times does not answer whether a compound inhibits growth, induces apoptosis, reaches a tumor, spares normal cells, or improves an in vivo outcome.

The candidate hierarchy follows this evidence boundary. Mocetinostat combines strict unseen-drug testing, low training-neighbor similarity, official measured reversal, and high rank stability and is therefore the core hypothesis. NCH-51 is secondary because it has measured support and stable ranks but residual scaffold exposure. TC-H-106 remains of interest, including because compounds in this chemical-development area have been investigated for central nervous system exposure [31], but four high-quality profiles and a 0.789 training neighbor do not justify principal-candidate status. RG2833 is prediction-only, and the Tianeptinaline/BG-1010 name-structure conflict prevents primary interpretation. This hierarchy replaces the original claim that TC-H-106, RG2833, and Tianeptinaline were equivalently and stably supported.

The class-level interpretation is deliberately conservative. HDAC inhibitors are established oncology agents and experimental tools [34–36]; recovering their transcriptional signatures does not constitute discovery of HDAC biology. Explicitly annotated class I HDAC structures were overrepresented at every fixed top-library revision cutoff under signed wTRS, but not under the co-primary Spearman reversal score. Signed wTRS changes with perturbational amplitude, whereas Spearman correlation is invariant to positive rescaling; the significant enrichment may therefore reflect both directional opposition and overall response magnitude. The useful result is narrower than a universal class claim: the screen identifies specific measured HDAC compounds and a signed-wTRS class signal while quantifying sensitivity to strict structure removal, alternate encoders, reversal

definition, and official-condition aggregation. Candidate-specific brain penetrance may be relevant to a later study of central nervous system disease or metastasis, but it is neither systematically evaluated here nor the basis of the revised core candidate.

Network and enrichment results support biological coherence without supplying independent target validation. The top genes were selected from the same disease-perturbation opposition used for ranking, so cell-cycle enrichment cannot be used twice as both discovery and validation. Describing them as reversal-associated genes resolves this circularity. The partial overlap with Entinostat and Vorinostat is consistent with related HDAC-class programs, while the sparser and less enriched TC-H-106 results reinforce its exploratory tier. Direct targets would require binding, occupancy, biochemical inhibition, or well-controlled perturbational genetics.

DepMap provides a separate but still indirect layer. The frozen 26Q1 analysis does not support a broad HDAC1 dependency narrative; HDAC3 is the dominant pan-cancer loss-of-function dependency. This is compatible with class I HDAC relevance but does not identify which isoform mediates the effect of Mocetinostat, NCH-51, or TC-H-106. Genetic knockout can also differ substantially from partial, time-limited, multi-isoform pharmacological inhibition. The prior non-significant GDSC surrogate comparison was removed rather than presented as support because Entinostat viability is not a valid direct validation of TC-H-106 reversal.

The docking controls similarly reduce rather than inflate the claim. Zinc-aware redocking recovered the crystallographic pose, demonstrating that the protocol can reproduce one known geometry under an appropriate potential. Yet the designed decoy scored well despite failing the intended zinc-binding-group geometry, and several compounds changed pose between HDAC2 and HDAC1. These findings illustrate why an affinity score or attractive rendering cannot establish direct binding. Docking remains useful for identifying structural uncertainties and planning biochemical experiments, not for validating a target.

Several limitations remain. First, both predictors use chemical structure only: cell identity, dose, and time are absent, so the output represents a structure-conditioned central tendency and leave-cell-line performance is not evidence of learned cell-specific response. Second, TCGA disease signatures are derived from bulk tumors, whereas LINCS profiles arise from cell lines under restricted doses and times. Tumor microenvironment and cellular-composition effects can therefore contribute to disease vectors [33]. Third, although the corrected annotation-defined HDAC holdout

is strict by exact structure, 30 compounds are a modest class test and some test scaffolds remain represented in training. Strict leave-drug candidate estimates were available only for Mocetinostat and PCI-24781. Fourth, class I HDAC enrichment was significant only under signed wTRS, which is sensitive to perturbational amplitude; it was absent under scale-invariant Spearman reversal, so response magnitude may contribute to the signed-score result. A magnitude-controlled sensitivity analysis was considered but not added post hoc because no unique adjustment had been prespecified; selecting among alternative normalization or matching schemes after observing the metric divergence would introduce an additional unplanned analytic choice. Fifth, the measured comparison uses held-out profiles from the same LINCS ecosystem rather than a fully independent platform; it is experimental transcriptomics but not an external viability dataset. A fully condition-matched non-HDAC comparator set was not recoverable from the archived official-condition outputs, so the eight-compound analysis establishes concordance but not preferential validation relative to alternative perturbagens; the available 11,445-compound all-profile ranks are reported only as an unmatched sensitivity analysis. Sixth, candidate profile counts and condition coverage are unequal despite hierarchical aggregation. Seventh, rank stability is descriptive and depends on the screened library and chosen disease signatures. Eighth, DepMap genetic loss, enrichment, and docking answer different questions from compound response. Finally, no new viability, biochemical, organoid, animal, or clinical experiment was performed.

These limitations define a focused experimental roadmap. Candidate testing should begin with the frozen tiers, include established HDAC positive controls and inactive or zinc-chelation controls, use multiple cancer and matched non-malignant models, prespecify dose and time, and measure both viability and post-treatment transcriptomics. Isoform-selective biochemical assays would determine whether the class-level signal maps to HDAC1, HDAC2, HDAC3, or a combination. Patient-derived organoids or co-culture systems could then assess whether the reversal pattern persists in models that better preserve microenvironmental context. These experiments are needed before any efficacy, selectivity, or translational claim.

5 Conclusions

Rigorous split design, leakage auditing, structural-proximity and applicability assessment, and

calibrated evidence layers provide an auditable basis for pan-cancer transcriptomic-reversal analysis and hypothesis-generating candidate prioritization. The strong conventional fingerprint MLP was at least as predictive and substantially more computationally efficient than the dual-stream model; the added representation complexity is not justified by the current benchmarks. Within this framework, predicted-measured LINCS concordance supports the eight evaluated compounds, signed wTRS—but not Spearman reversal—shows class I HDAC enrichment, and the evidence tiers retain Mocetinostat as core, NCH-51 as secondary, and TC-H-106 as exploratory. These results do not establish metric-independent enrichment, condition-specific response prediction, direct target engagement, or therapeutic efficacy. Separating generalization, chemical-space exposure, measured concordance, biological context, and structural sensitivity provides a more reliable basis for selecting future experiments.

List of abbreviations

AD4Zn: AutoDock4Zn; BBB: blood-brain barrier; BLCA: bladder urothelial carcinoma; CI: confidence interval; DepMap: Cancer Dependency Map; FDR: false discovery rate; GDC: Genomic Data Commons; GDSC: Genomics of Drug Sensitivity in Cancer; GO: Gene Ontology; HDAC: histone deacetylase; HiQ: official LINCS high-quality flag; KEGG: Kyoto Encyclopedia of Genes and Genomes; LINCS: Library of Integrated Network-Based Cellular Signatures; MLP: multilayer perceptron; PPI: protein-protein interaction; RMSD: root-mean-square deviation; TCGA: The Cancer Genome Atlas; TMM: trimmed mean of M-values; wTRS: weighted transcriptomic reversal score.

Declarations

Ethics approval and consent to participate

This study used publicly available, de-identified human transcriptomic datasets and publicly available pharmacological, dependency, and structural datasets. No new human participants, human tissue, animal experiments, or identifiable personal data were involved. Additional ethics approval and consent to participate were not required.

Consent for publication

Not applicable.

Availability of data and materials

The tumor disease signatures used in this study were derived from publicly available TCGA data accessed through the Genomic Data Commons (<https://portal.gdc.cancer.gov/>). Pharmacological perturbation profiles were obtained from LINCS L1000 (GEO accession GSE92742). DepMap Public 26Q1 was obtained from the DepMap portal, and structures 4LXZ and 4BKX were obtained from the RCSB Protein Data Bank. Source code, split-generation scripts, training and analysis scripts, machine-readable manifests, selected checksums, plotting data, and publicly redistributable processed outputs are available at <https://github.com/DCarchimonde/PanCancer-MultiModal-HDAC> on the `major-revision-2026` branch. The corrected analysis evidence is archived at commit `fefbf2f4fa7399983d4d4041dcb8e7e91b84d17f`, and the first-round submission snapshot at commit `0eb798f`. Repository documentation distinguishes original outputs, revision-regenerated analyses, retained derived evidence, large nonredistributed inputs or run artifacts, and intermediates that are no longer available. Reported values can be verified against archived machine-readable evidence and the submission documents recompiled from frozen table and figure assets; however, a self-contained bitwise rerun of every training and screening stage from the GitHub repository and current public inputs alone is not guaranteed because large GPU checkpoints and prediction arrays and specified analysis-time intermediates were not retained in the portable archive. Large primary public datasets are not redistributed because of file size and source-specific access terms. Stage-specific requirements and limitations are documented in `REPRODUCIBILITY.md`. Additional file 1 contains supplementary methods, tables, and figure notes; Additional file 2 contains the machine-readable revision tables.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

ST conceptualized the study, curated the data, performed the formal analysis and investigation, developed the software and methodology, validated the results, generated visualizations, and drafted and revised the manuscript. WZ contributed to methodology, supervision, project administration, and manuscript review and editing. SJ contributed to conceptualization, methodology, resources, funding acquisition, supervision, project administration, and manuscript review and editing. All authors read and approved the final manuscript.

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Authors' information

Not applicable.

List of additional files

Additional file 1 (.pdf): Supplementary material. Supplementary methods, Tables S1–S24, and Figures S1–S6 documenting dataset and split audits, repeated model evaluation, candidate identity and structural exposure, measured LINCS analyses, pathway and network results, DepMap context, docking controls, and reproducibility boundaries.

Additional file 2 (.xlsx): Machine-readable revision tables. A 58-sheet companion workbook containing the tabular evidence underlying the main and supplementary results, including dataset and split inventories, run-level and paired model metrics, candidate membership and identity, official-condition measured-profile coverage, enrichment and stability analyses, structural-neighbor audits, network and pathway results, DepMap summaries, docking controls, and the reproducibility inventory.

Use of large language models and AI-assisted technologies

During manuscript preparation, the authors used ChatGPT (OpenAI) for language editing, organization, and formatting support. No large language model was listed as an author, cited as a scientific source, used to generate experimental data, or used as a substitute for scientific interpretation. All numerical analyses, tables, and figures reported in the manuscript were generated by the versioned analysis code and verified against machine-readable outputs. The authors reviewed and edited the manuscript and take full responsibility for its content.

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